【例 6.123】(南开大学,2009 年) 设 P 为数域,T 为 $P^{n\times n}$ 上的线性变换,满足条件:对任何固定的 $A,B\in P^{n\times n}$,$T(AB)=T(A)T(B)$ 或 $T(AB)=T(B)T(A)$ 至少有一个成立. 证明:要么对所有的 $A,B\in P^{n\times n}$,$T(AB)=T(A)T(B)$,要么对所有的 $A,B\in P^{n\times n}$,$T(AB)=T(B)T(A)$.

Theorem: If a vector space is the union of two subspaces, then one of them equals the whole space

Proof. if $W=U\cup V$ and both are proper subspaces
choose $u\in U, v\in V$. Consider $u+v$
If $u+v\in U \to v\in U \to \times$
If $u+v\in V \to u\in V \to \times$
$u+v$ belongs to neither subspace
→ a contradiction

Now write $M=P^{n\times n}$. For $A\in M$
define $U_A=\{B\in M: T(AB)=T(A)T(B)\}$
$V_A=\{B\in M: T(AB)=T(B)T(A)\}$
both are subspaces of M
The hypothesis gives
$M=U_A\cup V_A$
Thus $U_A=M$ or $V_A=M$

Next, define

$U = \{A \in M : T(AB) = T(A)T(B) \text{ for every } B \in M\}$

$V = \{A \in M : T(AB) = T(B)T(A) \text{ for every } B \in M\}$

$M = U \cup V$

$\rightarrow U = M$ or $V = M$

Hence either

$T(AB) = T(A)T(B)$ for all $A, B \in M$

or

$T(AB) = T(B)T(A)$ for all $A, B \in M$

Invertibility and isomorphism

Invertible

one-to-one $\leftrightarrow$ onto

T is linear $\leftrightarrow$ T^{-1} is linear

$\text{rank}(T) = \dim(V) \leftrightarrow \text{nullity}(T) = 0$

$\Updownarrow$

$T: V \rightarrow W$ V is isomorphic to W

$\Updownarrow$

$\dim(V) = \dim(W)$

15. Let V and W be finite-dimensional vector spaces, and let $T: V \to W$ be a linear transformation. Suppose that β is a basis for V. Prove that T is an isomorphism if and only if $T(\beta)$ is a basis for W.

$\Rightarrow$ Suppose T is an isomorphism,
For $\forall w \in W$, $\exists v \in V$ i.e. $w = T(v)$
$\beta = \{v_1, \cdots, v_n\}$ is a basis, we can write
$$v = \sum_i a_i v_i$$
By linearity
$$w = T(v) = \sum_i a_i T(v_i)$$
Thus $T(\beta)$ spans W
Linear dependence:
suppose $\sum_i a_i T(v_i) = 0$
Then $T(\sum_i a_i v_i) = T(0)$
Injectivity gives $\sum_i a_i v_i = 0 \to$ every $a_i = 0$
$\to$ Q.E.D.

$\Leftarrow$: Suppose $T(\beta)$ is a basis for W
Every $w \in W$ can be written as
$$w = \sum_i a_i T(v_i) = T\left(\sum_i a_i v_i\right)$$
$\to T$ is surjective
Let $v = \sum_i b_i v_i$, $T(v) = \sum_i b_i T(v_i) = 0 \to$ every $b_i = 0$
$\to v = 0 \to \ker(T) = \{0\} \to T$ is injective

The change of coordinate matrix

$$[v]_\beta = Q[v]_{\beta'}$$

$\beta = (u_1, u_2)$ $\beta' = (v_1, v_2)$

Let $v = v_1$ $[v_1]_\beta = Q\begin{pmatrix}1\\0\end{pmatrix}$

Let $v = v_2$ $[v_2]_\beta = Q\begin{pmatrix}0\\1\end{pmatrix}$

$$Q = [[v_1]_\beta \quad [v_2]_\beta] = [I_V]^\beta_{\beta'}$$

2. For each of the following pairs of ordered bases β and β' for R^2, find the change of coordinate matrix that changes β'-coordinates into β-coordinates.

(a) $\beta = \{e_1, e_2\}$ and $\beta' = \{(a_1, a_2), (b_1, b_2)\}$

(b) $\beta = \{(-1, 3), (2, -1)\}$ and $\beta' = \{(0, 10), (5, 0)\}$

(a) $[I_V]^\beta_{\beta'} = [[(a_1, a_2)]_\beta \quad [(b_1, b_2)]_\beta] = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$

(b) $[I_V]^\beta_{\beta'} = [[(0, 10)]_\beta \quad [(5, 0)]_\beta] = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$

For linear operator T

$$[T]_{\beta'} = [ITI]_{\beta'} = [I]^{\beta'}_{\beta}[T]^{\beta}_{\beta}[I]^{\beta}_{\beta'} = Q^{-1}[T]_\beta Q$$

6. For each matrix A and ordered basis β, find $[\mathsf{L}_A]_\beta$. Also, find an invertible matrix Q such that $[\mathsf{L}_A]_\beta = Q^{-1}AQ$.

(a) $A = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$ and $\beta = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$

Let α be the standard basis, $A = [L_A]_\alpha$

$[L_A]_\beta = [I]_\alpha^\beta [L_A]_\beta^\beta [I]_\beta^\alpha$

$[I]_\beta^\alpha = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$

$[I]_\alpha^\beta = \left([(1,0)]_\beta \; [(0,1)]_\beta \right)$

$= \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$

$[L_A]_\beta = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$

$= \begin{pmatrix} 1 & 5 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 11 \\ -2 & -4 \end{pmatrix}$

Dual spaces

$V^* = \mathcal{L}(V, F)$ ← output is a scalar linear functional on V

$\dim(V^*) = \dim(V)$

i-th coordinate function $f_i(x) = a_i$ where $x = \sum_i a_i x_i$

$\beta^* = \{f_1, \cdots, f_n\}$ is an ordered basis for V^*

For any $f \in V^*$ $\quad f = \sum_{i=1}^{n} f(x_i) f_i$

1. Label the following statements as true or false. Assume that all vector spaces are finite-dimensional.

(a) Every linear transformation is a linear functional. ✗ (scalar)

(b) A linear functional defined on a field may be represented as a 1×1 matrix. ✓

(c) Every vector space is isomorphic to its dual space. ✓

(d) Every vector space is the dual of some other vector space. ✓

(e) If T is an isomorphism from V onto V^* and β is a finite ordered basis for V, then $T(\beta) = \beta^*$. ✗ (coefficient $e_i^*(e_j) = \delta_{ij}$)

(f) If T is a linear transformation from V to W, then the domain of $(T^t)^t$ is V^{**}. ✓

(g) If V is isomorphic to W, then V^* is isomorphic to W^*. ✓ (dim)

(h) The derivative of a function may be considered as a linear functional on the vector space of differentiable functions. ✗ (still a function, not a scalar)

(d) $V \to W^* = V^{**}$

V^*: $f(x,y) = x$ $\quad g(x,y) = y \cdots$

$W^* = V^{**}$: Denote

$$J(v)(f) = f(v)$$

$J(v)(f) = f(x,y) = x$ $\quad J(v)(g) = g(x,y) = y \cdots$

Thus $J: V \to V^{**}$, $\dim(V^{**}) = \dim(V^*) = \dim(V)$

J is also injective $\longrightarrow$ J is isomorphism

(f) $T: V \to W$ $\quad T^t: W^* \to V^*$ $\quad (T^t)^t: V^{**} \to W^{**}$

5. Let $\mathsf{V} = \mathsf{P}_1(R)$, and, for $p(x) \in \mathsf{V}$, define $\mathsf{f}_1, \mathsf{f}_2 \in \mathsf{V}^*$ by

$$\mathsf{f}_1(p(x)) = \int_0^1 p(t)\,dt \quad \text{and} \quad \mathsf{f}_2(p(x)) = \int_0^2 p(t)\,dt.$$

Prove that $\{\mathsf{f}_1, \mathsf{f}_2\}$ is a basis for V^*, and find a basis for V for which it is the dual basis.

Assume that $p(x) = a + bx$

$f_1(p(x)) = \int_0^1 (a+bx)dx = a + \frac{b}{2}$

$f_2(p(x)) = \int_0^2 (a+bx)dx = 2a + 2b$

Linear independence: $c_1 f_1 + c_2 f_2 = 0$

$$\begin{cases} c_1 + 2c_2 = 0 \\ \frac{c_1}{2} + 2c_2 = 0 \end{cases} \rightarrow \begin{cases} c_1 = 0 \\ c_2 = 0 \end{cases}$$

Dual basis

$a + bx = (a + \frac{b}{2})(m + nx) + (2a + 2b)(p + qx)$

$m(a + \frac{b}{2}) + p(2a + 2b) = a$

$$\begin{cases} m + 2p = 1 \\ \frac{m}{2} + 2p = 0 \end{cases} \rightarrow \begin{cases} m = 2 \\ p = -\frac{1}{2} \end{cases}$$

$n(a + \frac{b}{2}) + q(2a + 2b) = b$

$$\begin{cases} n + 2q = 0 \\ \frac{n}{2} + 2q = 1 \end{cases} \rightarrow \begin{cases} n = -2 \\ q = 1 \end{cases}$$

$a + bx = (a + \frac{b}{2})(2 - 2x) + (2a + 2b)(-\frac{1}{2} + x)$

$\{2 - 2x, -\frac{1}{2} + x\}$ is the dual basis